

Chapter 16: Abstract Vector Spaces Worksheet

1. Are each of the following statements true or false?

- (a) The set of 2×2 matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ such that $abcd = 0$ is closed under addition.
- (b) The set of polynomials of degree exactly 3 is closed under addition.
- (c) In the vector space $M_{2 \times 3}$ with standard matrix addition and scalar multiplication, the zero vector is $\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.
- (d) The set of noninvertible 3×3 matrices is closed under scalar multiplication.

2. Consider P_2 , the vector space of polynomials of degree 2 or less. Find, if possible, a, b, c such that $x^2 + 9x - 11 = a\mathbf{u} + b\mathbf{v} + c\mathbf{w}$ for $\mathbf{u} = x^2 - 1$, $\mathbf{v} = x^2 - 2x + 3$, $\mathbf{w} = 2x^2 + 3x$.

3. Let V be the set of 2×2 matrices with standard matrix addition; however, define scalar multiplication $\odot$ by $c \odot A = cA^T$. For each scalar multiplication axiom (S1)-(S5), determine if $\odot$ satisfies the axiom. If the axiom is satisfied, prove it (remember to use variables). If the axiom is not satisfied, give an explicit counterexample to demonstrate why. (Hint: the axioms are listed in the next problem).

4. Let V be the set of all positive real numbers; define vector addition $\oplus$ by $u \oplus v = uv$ ($\oplus$ is ordinary multiplication) and define scalar multiplication $\odot$ by $c \odot v = v^c$. Prove that V is a vector space. The axioms are provided below for your convenience.

(A1) If $\mathbf{u}$ and $\mathbf{v}$ are in V , then $\mathbf{u} \oplus \mathbf{v}$ is in V .

(A2) $\mathbf{u} \oplus \mathbf{v} = \mathbf{v} \oplus \mathbf{u}$ for all $\mathbf{u}, \mathbf{v}$ in V .

(A3) $\mathbf{u} \oplus (\mathbf{v} \oplus \mathbf{w}) = (\mathbf{u} \oplus \mathbf{v}) \oplus \mathbf{w}$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V .

(A4) There exists an element $\mathbf{0}$ in V such that $\mathbf{u} \oplus \mathbf{0} = \mathbf{0} \oplus \mathbf{u} = \mathbf{u}$ for all $\mathbf{u}$ in V .

(A5) For each $\mathbf{u}$ in V there exists an element $-\mathbf{u}$ in V such that $\mathbf{u} \oplus -\mathbf{u} = -\mathbf{u} \oplus \mathbf{u} = \mathbf{0}$.

(S1) If $\mathbf{v}$ is in V and r is in $\mathbb{R}$, then $r \odot \mathbf{v}$ is in V .

(S2) $r \odot (\mathbf{u} \oplus \mathbf{v}) = r \odot \mathbf{u} \oplus r \odot \mathbf{v}$ for all $\mathbf{u}, \mathbf{v}$ in V and r in $\mathbb{R}$.

(S3) $(r + s) \odot \mathbf{v} = r \odot \mathbf{v} \oplus s \odot \mathbf{v}$ for all $\mathbf{v}$ in V and r, s in $\mathbb{R}$.

(S4) $r \odot (s \odot \mathbf{v}) = (rs) \odot \mathbf{v}$ for all $\mathbf{v}$ in V and r, s in $\mathbb{R}$.

(S5) $1 \odot \mathbf{v} = \mathbf{v}$ for all $\mathbf{v}$ in V .

5. Consider each given operation(s) on $\mathbb{R}^2$ (written as ordered pairs) and each axiom. If the axiom is satisfied by the operation, prove it (reminder: use variables). If it's not satisfied, give an explicit counterexample to demonstrate why. (Hint: the axioms are listed in the previous problem).

- (a) Standard addition, scalar multiplication defined as $c \odot (x, y) = (cx, -cy)$; consider S2.
- (b) Standard addition, scalar multiplication defined as $c \odot (x, y) = (cx, -cy)$; consider S4.
- (c) Standard addition, scalar multiplication defined as $c \odot (x, y) = (x, y)$; consider S3.
- (d) Addition defined as $(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 + y_2 + 1)$, scalar multiplication defined as $c \odot (x, y) = (cx, cy + c - 1)$; consider A2.
- (e) Addition defined as $(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 + y_2 + 1)$, scalar multiplication defined as $c \odot (x, y) = (cx, cy + c - 1)$; consider S4.
- (f) Addition defined as $(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 + y_2 + 1)$, scalar multiplication defined as $c \odot (x, y) = (cx, cy + c - 1)$; consider S5.